

DAVIDE PALLESCHI

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SUMMARY

Applied AI Researcher specializing in Large Language Models and Multi-Agent Systems. I excel at prototyping and scaling advanced AI architectures, from **RAG** to hierarchical **agent orchestration**. By integrating research-focused experimentation with professional software engineering principles, I deliver reliable AI solutions that solve complex automation tasks through rigorous benchmarking and model optimization.

EXPERIENCE

European Court of Auditors Artificial Intelligence Engineer

Luxembourg
04/2026 – Present

- Integrating LLM-based solutions and multi-agent systems to enhance institutional audit processes.
- Focusing on technical robustness and alignment with the EU AI Act framework.

DeepPlants AI Researcher

Rome, Italy
09/2024 – 02/2026

- Designed autonomous multi-agent architectures (**LangGraph**) to orchestrate complex workflows.
- Fine-tuned LLMs (**LoRA/PEFT**) and engineered advanced **RAG** systems for highly reliable document analysis and retrieval.
- Implemented automated benchmarking pipelines to evaluate generative model performance and output reliability.

Sapienza S.r.l. Software Engineer

Rome, Italy
04/2018 – 07/2024

- Architected scalable Python automation systems and robust API pipelines, streamlining core business workflows.
- Maintained and optimized backend infrastructure and databases to ensure data integrity and high availability.

EDUCATION

Sapienza University of Rome — *M.Sc., AI and Robotics*

09/2020 – 03/2024

- Grade: 102/110 • Erasmus: Polytechnic University of Valencia (2022)
- Thesis: *Privacy-Oriented Object Detection (low-resolution video, YOLOv8)*

University of Rome Tor Vergata — *B.Sc., Computer Engineering*

09/2015 – 08/2020

- Grade: 101/110 • Erasmus: Imperial College London (2018–2019)

SKILLS

Agentic Systems: Multi-agent orchestration, Hierarchical agents, LangGraph, LangChain

LLM Research & Fine-Tuning: PEFT/LoRA Fine-tuning, Hugging Face Transformers, PyTorch

Software Engineering & Stack: Python, FastAPI, Docker, vLLM, Database/SQL, Software Architecture

Languages: Italian (Native), English (Professional)

I hereby authorize the processing of my personal data in accordance with EU Regulation 2016/679 (General Data Protection Regulation) for the purpose of recruitment and selection.